

# Computational hwk #1

Nicole James

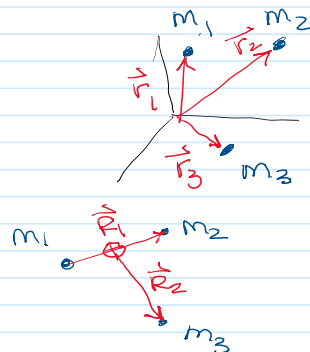
3/20/26

1) given: 3BP EOM in Jacobi coord

$$m_1 + m_2 + m_3 = 1 \quad G = 1$$

input: masses,  $\vec{R}_0$ ,  $\vec{V}_0$

output: time series of  $\vec{r}$  &  $\vec{v}$



$$\vec{r}_1 = \vec{r}_{12} = \vec{r}_2 - \vec{r}_1$$

$$\vec{r}_2 = \frac{1}{m_1 + m_2} [m_1(\vec{r}_3 - \vec{r}_1) + m_2(\vec{r}_3 - \vec{r}_2)]$$

$$\vec{r}_N = \vec{r}_c$$

$$\vec{r}_{N-1} = \frac{1}{\sum_{i=1}^{N-1} m_i} \sum_{i=1}^{N-1} m_i \vec{r}_{iN}$$

$$\vec{r}_{12} = \vec{r}_1$$

$$\vec{r}_{23} = \vec{r}_2 - \frac{m_1}{m_1 + m_2} \vec{r}_1$$

$$\vec{r}_{13} = \vec{r}_2 - \frac{m_2}{m_1 + m_2} \vec{r}_1$$

$$M_i = \sum_{j=1}^N m_j$$

$$T = \sum_{i=1}^{N-1} M_{i+1} \frac{M_{i+1} m_i}{M_i} \dot{\vec{r}}_i \cdot \dot{\vec{r}}_i + \frac{1}{2} M_N \dot{\vec{r}}_N \cdot \dot{\vec{r}}_N$$

$$H = \sum_{i=1}^{N-1} \frac{M_{i+1} m_i}{M_i} \dot{\vec{r}}_i \times \dot{\vec{r}}_i + M_N \dot{\vec{r}}_N \times \dot{\vec{r}}_N$$

$$I_p = \sum_{i=1}^{N-1} \frac{M_{i+1} m_i}{M_i} R_i^2 + M_N R_N^2$$

$$\vec{y} = \begin{bmatrix} \vec{r} \\ \dot{\vec{r}} \end{bmatrix} \quad \frac{d\vec{y}}{dt} = \dot{\vec{y}} = \begin{bmatrix} \dot{\vec{r}} \\ \ddot{\vec{r}} \end{bmatrix}$$

$$U = - \sum_{i=1}^{N-1} \sum_{j=i+1}^N \frac{G m_i m_j}{|\vec{r}_{ij}|}$$

Lagrangian Eqs  $\Rightarrow$  EOM

$$\frac{\partial \vec{r}_i}{\partial \vec{r}_j} = \begin{cases} 0 & j \geq i \\ 1 & j = i-1 \\ \frac{m_{j+1}}{m_{j+1} + 1} & j \leq i-2 \end{cases}$$

$$\frac{\partial U}{\partial \vec{r}_k} = -G \sum_{l=1}^{N-1} \sum_{j=l+1}^N \frac{m_l m_j}{r_{lj}^3} \vec{r}_{lj} \cdot \frac{\partial \vec{r}_{lj}}{\partial \vec{r}_k}$$

$$L = T - U$$

$$\frac{\partial L}{\partial \vec{r}_j} = \frac{M_{j+1} m_j}{M_j} \ddot{\vec{r}}_j$$

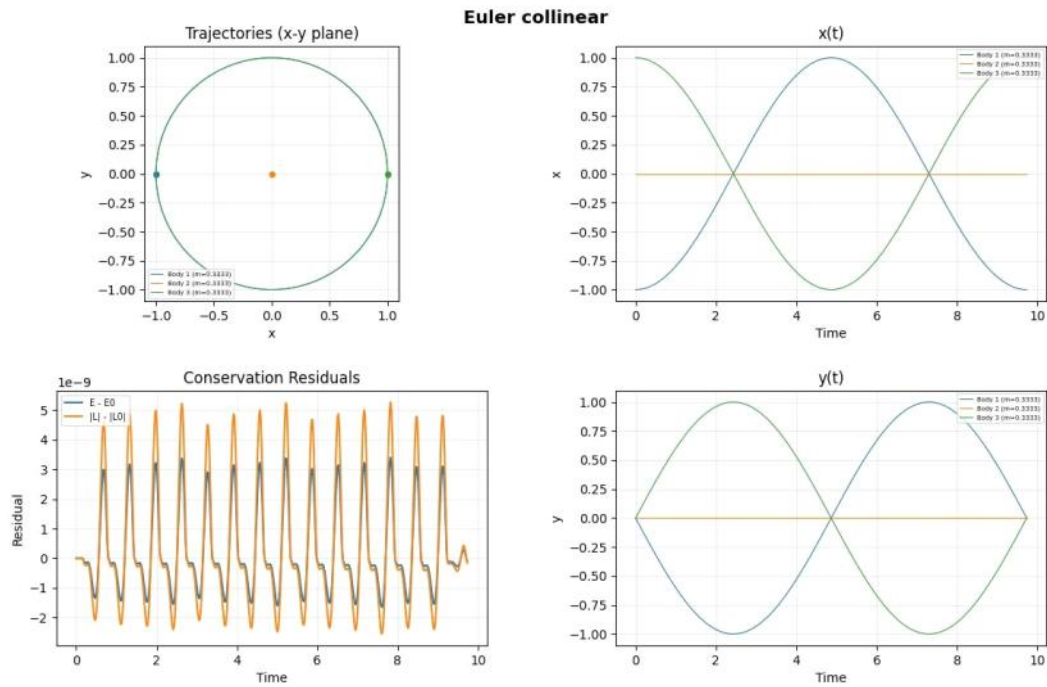
$$\frac{d}{dt} \left( \frac{\partial L}{\partial \dot{\vec{r}}_j} \right) = \frac{M_{j+1} m_j}{M_j} \ddot{\vec{r}}_j$$

1a) rel tol =  $1 \times 10^{-8}$

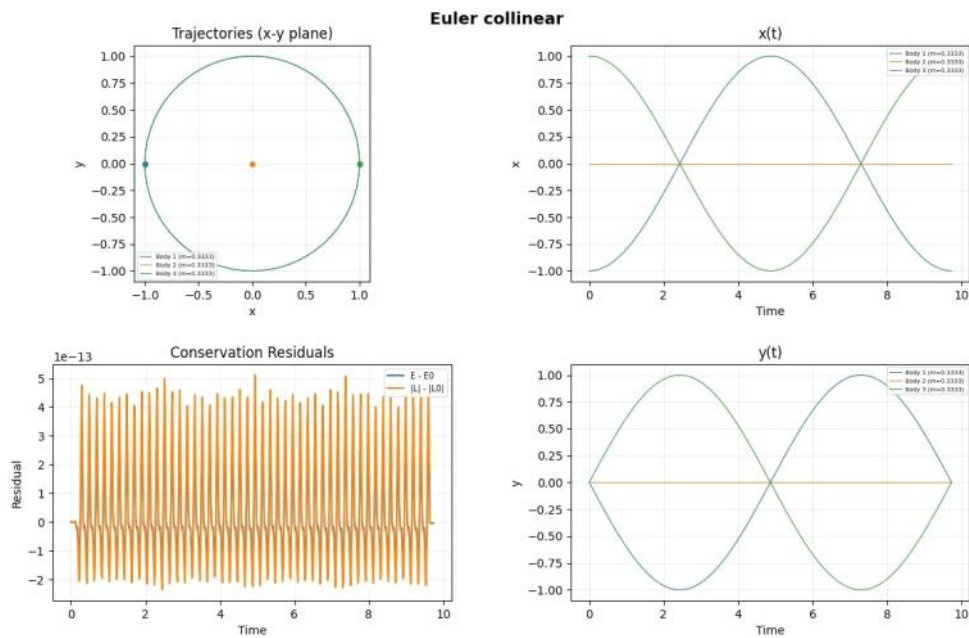
abs tol =  $1 \times 10^{-10}$

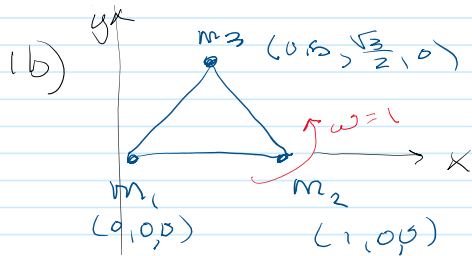
(using Euler  
const mass conf(q))

1a)  $\text{rel tol} = 1 \times 10^{-8}$  (using Euler  
 $\text{abs tol} = 1 \times 10^{-10}$  const mass config)



$\text{rel tol} = 1 \times 10^{-12}$   
 $\text{abs tol} = 1 \times 10^{-14}$



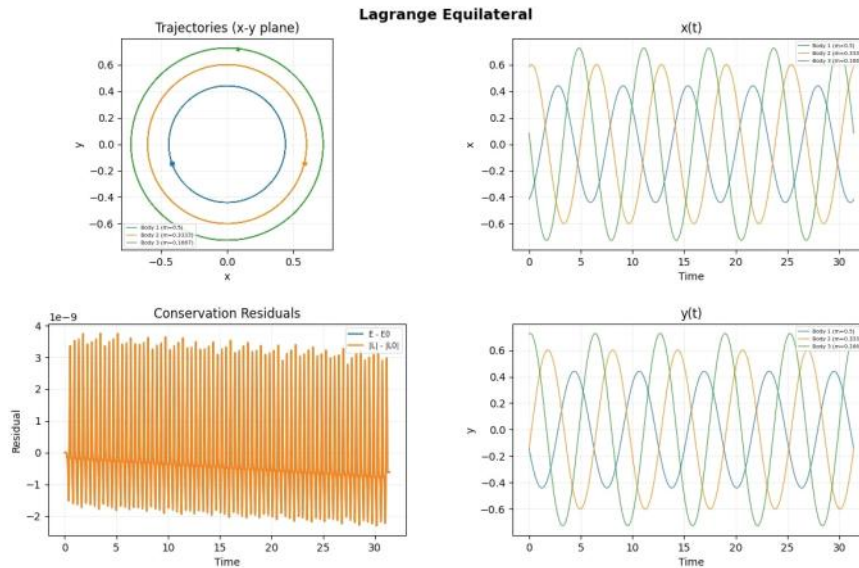


$$m_1 = \frac{1}{2} \quad m_2 = \frac{1}{3} \quad m_3 = \frac{1}{6}$$

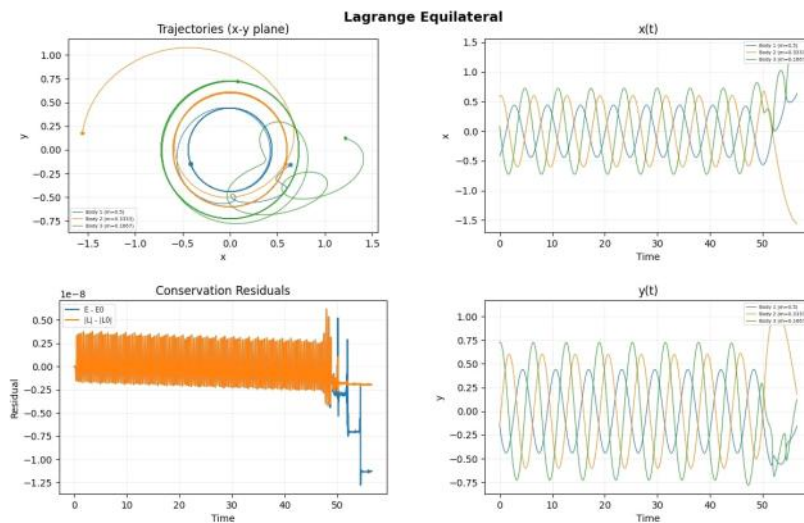
$$\omega = 1 \quad \vec{v} = \omega \vec{r}$$

$$\text{period} = 2\pi / \omega$$

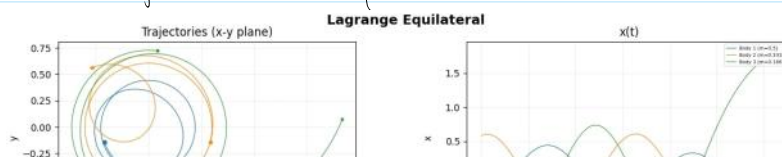
The orbit stays stable through about 8 periods

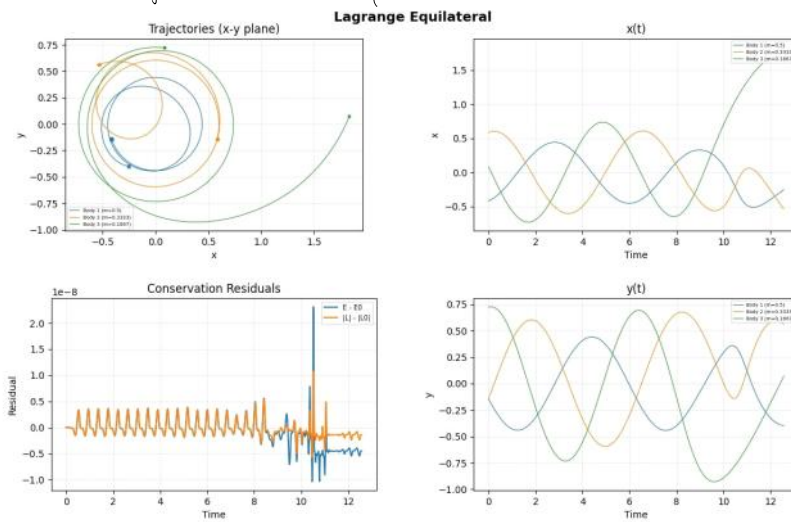


During the 9th period, it becomes quite chaotic

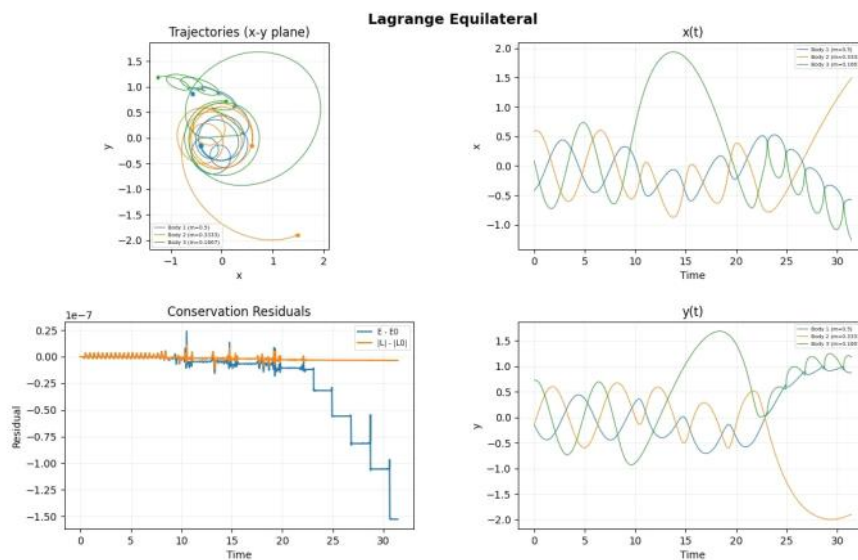


Adding a small perturbation to a single body (body 2) results in a large difference in each body's trajectory, this after 2 periods





This is after 5 periods:



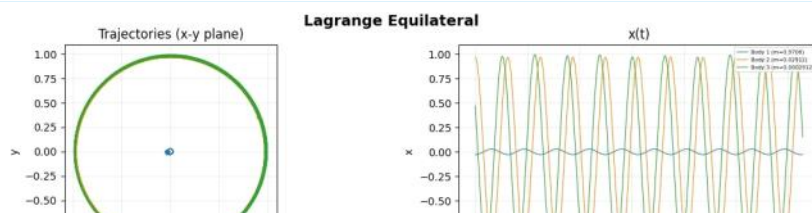
c) repeating (b) but with masses that meet the Routh stability criterion

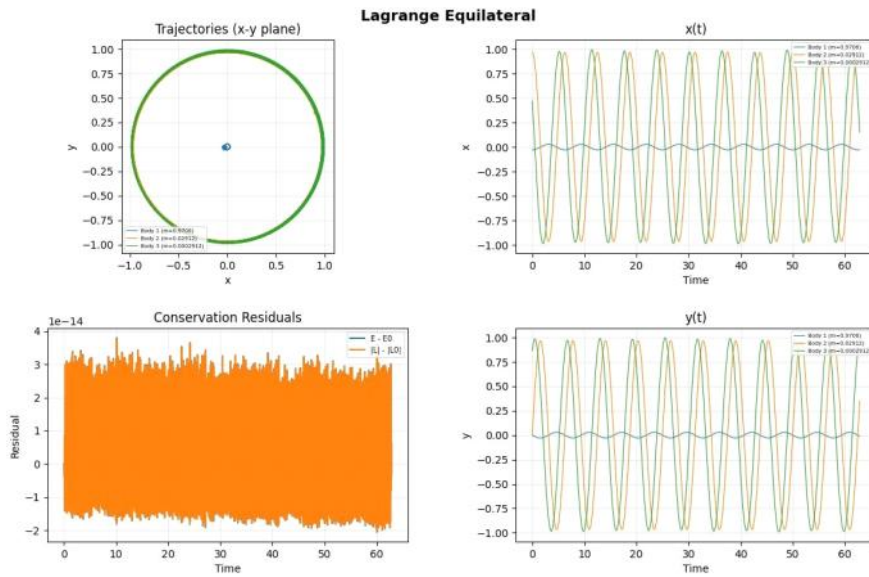
$$m_1 = \frac{1}{m_1 + m_2 + m_3}$$

$$m_2 = \frac{0.03}{m_1 + m_2 + m_3}$$

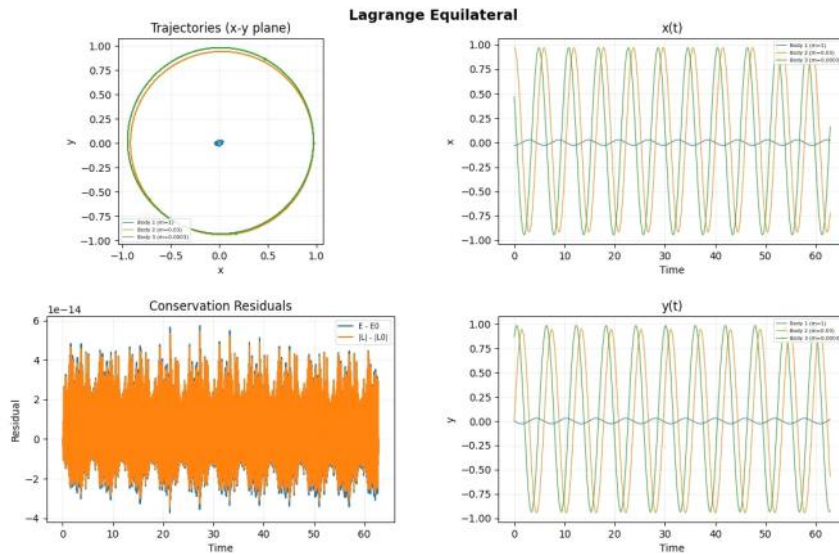
$$m_3 = \frac{0.0003}{m_1 + m_2 + m_3}$$

They are stable & stay in equilateral triangle config





Adding 0.001 to  $r_2$  results in small deviations but system is still stable & in equilateral triangle config.



The system stays stable thru the 1<sup>st</sup> 10 orbits until I perturb  $m_2$  by 0.007. Then the system config moves away from triangular config.

### Lagrange Equilateral

